# Imbalance-Aware and Semantic-Structured Learning for Multi-Label Image Tagging on Web-Scale Data: A Literature Survey

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**Abstract**—Multi-label image tagging plays a pivotal role in diverse applications such as image retrieval, autonomous systems, and scene understanding; however, the task remains challenging due to extreme data imbalance, complex label semantics, and the inherent intricacies of data fusion. This literature review presents a comprehensive analysis of the problem definition, benchmark datasets, and the techniques designed to address these challenges, alongside the metrics required for rigorous evaluation. Firstly, we provide a formal problem definition and systematically examine five of the most widely used datasets in multi-label learning, categorizing them by label cardinality to delineate how scale influences model behavior. Secondly, by evaluating current strategies across imbalance-handling, semantic correlation modeling, and multi-source fusion, we identify critical research gaps—specifically the tendency of existing models to treat these interconnected challenges in silos. Finally, in response, we propose a research direction for our project that advocates for a unified framework, synergizing feature-rich fusion with bias-aware, high-order semantic dependencies to achieve robust performance in large-scale, real-world tagging environments.

**Index Terms**—Multi-label image tagging, data fusion, long-tailed learning, label correlation.

## 1 Introduction

Multi-label image tagging has become a fundamental task in computer vision, aiming to assign multiple semantic labels to a single image [1]. Compared with traditional single-label classification, it better reflects the complexity of real-world visual data, where multiple objects and concepts coexist. This capability is essential for a wide range of applications, including image retrieval, recommendation systems, and visual question answering. In particular, models must simultaneously deal with highly imbalanced label distributions, capture complex semantic dependencies among labels, and effectively integrate heterogeneous visual features [2]. Therefore, addressing data imbalance, semantic co-occurrence, and data fusion has become a critical challenge in multi-label image tagging.

Among these challenges, data imbalance is one of the most critical issues. In multi-label settings, each instance is associated with only a few positive labels but a large number of negative ones, resulting in highly asymmetric learning signals [3]. Moreover, large-scale datasets typically exhibit a long-tail distribution, where a small number of frequent labels dominate the training process while rare labels are severely underrepresented [4]. This imbalance leads to biased optimization, where models tend to favor frequent labels and fail to learn discriminative representations for rare concepts. To address this, a variety of data-level and algorithm-level strategies have been proposed such as oversampling, undersampling and loss function [3]. However, these approaches primarily operate at the optimization level. They adjust learning dynamics but do not fundamentally resolve the issue of weak feature representations or the lack of structural understanding among labels.

Beside this issue, labels are often treated independently, ignoring the potential semantic relationships between them. In practice, labels rarely appear in isolation; instead, they form structured co-occurrence patterns. Certain rare labels may strongly depend on the presence of more common contextual labels. To model such dependencies, second-order and high-order strategies existed. Second-order methods solve pairwise label correlations and are widely applied in large-scale settings. High-order strategies aim to capture more complex dependencies in multiple labels. These methods provide stronger representational capacity and can better capture semantic structures beyond simple pairwise relations. However, both strategies face critical limitations under long-tail distributions. Second-order methods are inherently limited in their ability to model complex semantic interactions. Meanwhile, high-order models often suffer from scalability issues and representation bias. So effectively modeling semantic dependencies in multi-label learning requires not only expressive structures but also robustness to extreme data imbalance is a big challenge.

Beyond label-level modeling, effective multi-label tagging also depends on the quality of visual representations. A single feature source is often insufficient to capture the rich and diverse semantics present in complex images. As a result, feature fusion has become a key component in modern approaches. Recent methods employ attention mechanisms or transformer-based architectures to integrate multi-level or multi-modal features [5]. While these techniques improve representation capacity, they still face limitations in long-tail scenarios [4]. In particular, fusion mechanisms are often biased toward dominant visual patterns associated with frequent labels, providing limited support for rare concepts. Moreover, many existing interaction modules remain relatively shallow, making it difficult to capture higher-order semantic relationships and complex spatial dependencies [6]. As a result, even with advanced architectures, the learned representations may still be insufficient for robust multi-label prediction under real-world conditions.

Despite significant progress, current approaches typically address data imbalance, semantic dependency, and feature fusion in isolation. This fragmented design limits their effectiveness, as improvements in one aspect do not necessarily translate into overall performance gains [7]. In particular, even with imbalance-aware loss functions, weak feature representations can hinder rare label prediction. Similarly, semantic modeling without considering imbalance may amplify biases toward frequent labels. Therefore, these challenges should not be treated independently but rather as interconnected components of a unified problem. Motivated by this observation, this review aims to provide a comprehensive perspective on multi-label image tagging by jointly analyzing imbalance-aware learning, semantic-based modeling, and deep feature fusion. The structure of this literature review is summarized in Fig. 1

## 2 Related Works

### 2.1 Multi-label image tagging

#### 2.1.1 Definition

Multi-Label Image Tagging (MLIT) aims to associate a single image with a set of semantic labels, reflecting the inherent complexity of real-world visual scenes where multiple objects, actions, and concepts coexist. Unlike traditional single-label classification, which assumes category mutual exclusivity and restricts an image to an isolated class, MLIT treats each instance as a point in a high-dimensional binary space.

Formally, let $\mathcal{X} \in \mathbb{R}^{H \times W \times 3}$ represent the input image space and $\mathcal{L} = \{l_1, l_2, \ldots, l_C\}$ be a predefined label space consisting of $C$ distinct categories. In the MLIT framework, for a given image $\mathbf{x} \in \mathcal{X}$, the ground-truth annotation is explicitly defined as a binary vector $\mathbf{y} \in \{0, 1\}^C$, where:

$$y_c = \begin{cases} 1, & \text{if label } l_c \text{ is relevant to } \mathbf{x} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

This formulation effectively separates the label vocabulary into relevant (active) and irrelevant (inactive) groups based on the visual content.

The primary objective of the tagging task is to learn a mapping function $f : \mathcal{X} \to \mathbb{R}^C$, typically parameterized by a deep neural network $\Theta$, that produces a continuous confidence score vector $\hat{\mathbf{y}}$:

$$\hat{\mathbf{y}} = f(\mathbf{x}; \Theta) = [\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_C]^\top \tag{2}$$

where $\hat{y}_c \in [0, 1]$ represents the predicted probability of the presence of the $c$-th label in the image. To obtain the final discrete label assignments, a thresholding or ranking strategy is subsequently applied to $\hat{\mathbf{y}}$.

Mathematically, the learning process for MLIT involves minimizing the empirical risk over a dataset $\mathcal{D}$ using a multi-label loss function $\mathcal{J}$ (e.g., binary cross-entropy):

$$\Theta^* = \arg\min_{\Theta} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}} \mathcal{J}(\mathbf{y}, f(\mathbf{x}; \Theta)) + \lambda R(\Theta) \tag{3}$$

where $R(\Theta)$ denotes the regularization term to prevent overfitting in large-scale label spaces.

Rather than treating $\mathcal{L}$ as a collection of independent targets, modern MLIT frameworks focus on capturing the joint probability of labels to account for strong co-occurrence patterns and semantic dependencies (e.g., "beach" and "sand"). This shift from independent binary classification to structured prediction is essential for resolving visual ambiguities.

However, as the label cardinality $C$ increases, MLIT faces several inherent challenges. In particular, large-scale image datasets typically exhibit extreme class imbalance, where label distributions follow a long-tailed pattern. In addition, the heterogeneity of visual features further increases the difficulty of effective representation learning. These challenges motivate the need for the advanced data modeling and fusion strategies discussed in the subsequent sections.

#### 2.1.2 Multi-label image dataset

In the current landscape of computer vision, a vast array of datasets has been developed to advance the field of multi-label image classification. These datasets are curated from diverse sources—ranging from controlled environments to unconstrained web imagery—and are annotated by either crowdsourced users or domain experts. In this review, we focus on five of the most widely adopted benchmarks: PASCAL VOC 2007/2012 [8], NUS-WIDE [9], MS-COCO (COCO-80) [10], VG-500 [11], and News-500 [12]. To provide a structured analysis, we categorize these datasets into three distinct tiers based on their label cardinality:

***Low-Cardinality Tier ($\leq$ 30 Classes)*** The first tier, referred to as low-cardinality, consists of datasets with a limited number of labels, typically under 30 classes. These benchmarks are designed to evaluate the most fundamental object recognition capabilities of a model. The most prominent dataset within this group is PASCAL VOC 2007/2012. Specifically, PASCAL VOC 2007 contains a trainval set of 5,011 images and a test set of 4,952 images, while the 2012 version expands the scale to 11,540 images for training/validation and 10,991 for testing. The dataset covers only 20 categories, which are organized into four broad super-classes: Vehicle, Household, Animal, and Person. Due to this narrow scope, PASCAL VOC primarily focuses on common, isolated objects with sparse co-occurrence patterns. Consequently, the dataset does not exhibit complex or high-order label dependencies. Therefore, while it remains a stable benchmark for baseline performance, PASCAL VOC is less effective for evaluating modern architectures that rely heavily on label correlation modeling. It is most suited for approaches that treat labels as independent entities rather than those aiming to exploit deep semantic interactions.

***Mid-Cardinality Tier (30–100 Classes)*** The mid-cardinality tier encompasses datasets that bridge the gap between basic recognition and complex scene understanding. Unlike the previous tier, these benchmarks do not only expand the number of categories but also introduce finer sub-classifications (e.g., diversifying the 'person' category into specific roles like 'police' or 'mailman') and, more importantly, incorporate environmental context alongside physical objects.

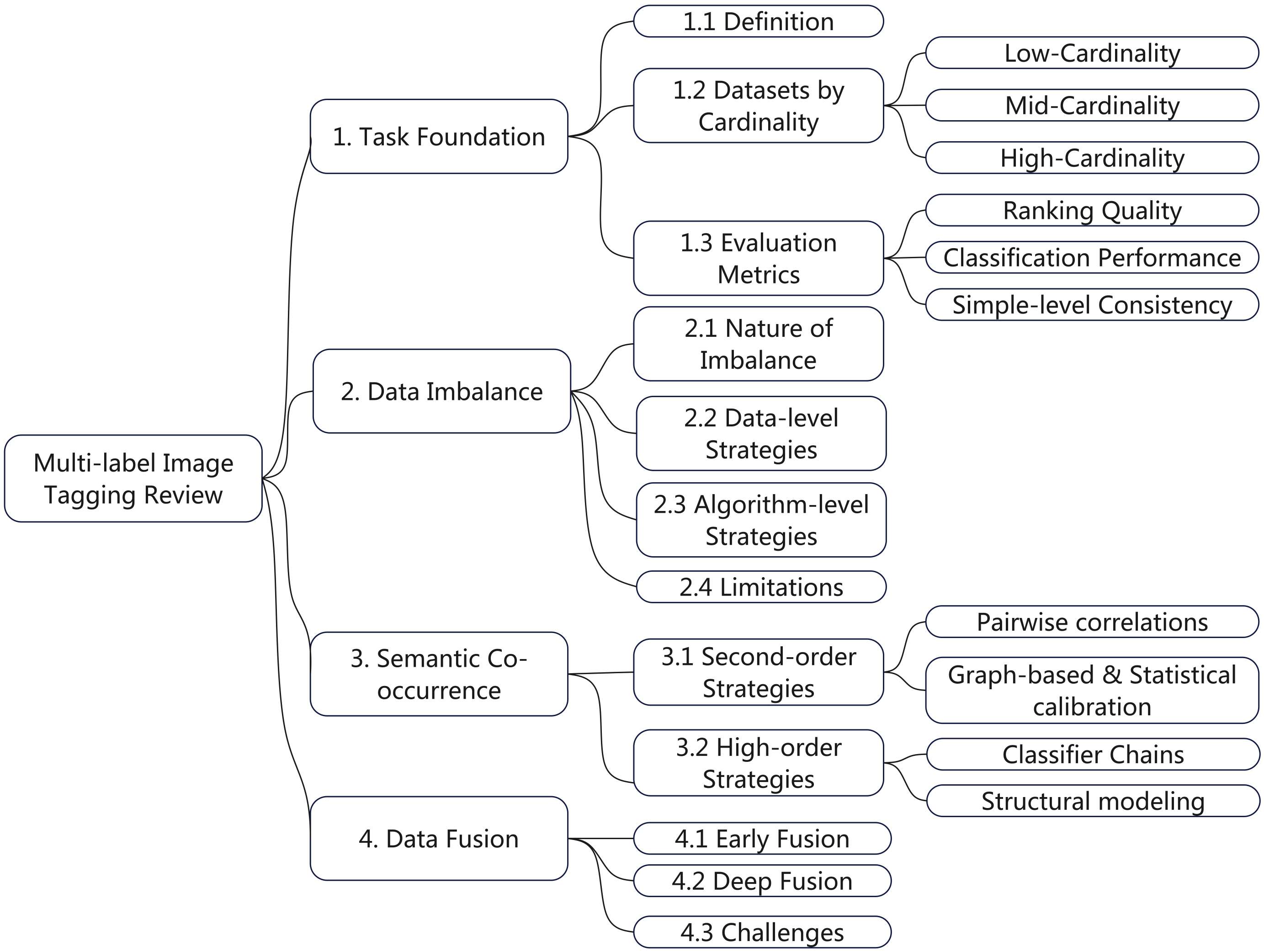


Fig. 1. The overview of this survey

MS-COCO (COCO-80) is a primary example, featuring 80 categories that span a broad spectrum from common entities such as person, bicycle, and car to more specialized or rare objects like surfboard and skis. The core strength of COCO lies in its focus on ”objects in context,” providing a rich ground for models to learn spatial co-occurrences. Furthermore, NUS-WIDE significantly scales up this complexity with 269,648 images and 81 curated categories. These labels are organized into six distinct super-classes: Event/Activity, Program, Scene/Location, Object, People, and Graphics. By including non-object tags like scenes and activities, NUS-WIDE offers a more diverse and heterogeneous semantic landscape. Consequently, these datasets are not only more computationally demanding but also represent a higher level of diversity, challenging models to capture both object-to-object and object-to-scene dependencies.

***High-Cardinality Tier (≥ 100 Classes)*** The high-cardinality tier refers to datasets containing hundreds or even thousands of categories. In this tier, the label space is not only significantly expanded in volume but also characterized by intricate and dense inter-dependencies. Beyond a simple increase in vocabulary, the categories at this level extend into fine-grained attributes, moving past mere object and scene recognition to encompass abstract concepts and actions.

Prominent examples of this tier include VG-500 and News-500. In these benchmarks, the categories go beyond physical entities to include high-level semantic concepts such as anti, celebrate, election, and feeding. This semantic expansion aligns the datasets more closely with complex, real-world contexts, where an image is often defined by the interplay of objects, actions, and social themes.

However, this increased proximity to real-world complexity renders prediction significantly more difficult. These large-scale datasets currently pose a formidable challenge in the field of multi-label image tagging and classification, as models must not only identify diverse visual cues but also reason across a highly complex and often long-tailed semantic landscape.

#### *2.1.3 Evaluation metrics*

No single metric is sufficient to fully characterize performance in multi-label image tagging. A comprehensive evaluation should instead consider ranking quality, label-wise classification behavior, and sample-level prediction consistency. This need is further amplified by practical challenges such as long-tailed label distributions, semantic co-occurrence among labels, and the increasing use of multi-feature fusion. Accordingly, this section reviews the evaluation metrics most commonly adopted in the literature on multi-label image tagging.

***Mean Average Precision (mAP)*** evaluates the ranking quality of predicted labels by measuring whether relevant labels are assigned higher confidence scores than irrelevant ones. For each label $l$, the Average Precision (AP) is computed from the precision-recall curve as

$$AP_l = \sum_{k=1}^{n} P_l(k)\,\Delta R_l(k) \tag{4}$$

where $P_l(k)$ denotes the precision at the $k$-th threshold or ranking position for label $l$, $\Delta R_l(k)$ denotes the change in recall at that position, and $n$ is the total number of prediction thresholds or ranked instances considered. The mean Average Precision is then obtained by averaging over all labels:

$$mAP = \frac{1}{L}\sum_{l=1}^{L} AP_l \tag{5}$$

where $L$ is the total number of labels and $AP_l$ is the Average Precision of the $l$-th label.

Larger mAP values indicate that relevant labels are more consistently ranked ahead of irrelevant ones, thereby reflecting stronger ranking performance. Lower values, in contrast, suggest weaker separability in the model's confidence ordering.

***F1-micro and F1-macro*** are label-based metrics used to evaluate multi-label classification performance from different perspectives. The micro-F1 score aggregates classification outcomes over all labels:

$$F1_{micro} = \frac{2TP}{2TP + FP + FN} \tag{6}$$

where $TP$ denotes the total number of true positives across all labels, $FP$ denotes the total number of false positives, and $FN$ denotes the total number of false negatives.

The macro-F1 score first computes the F1 score for each label independently and then averages them:

$$F1_{macro} = \frac{1}{L}\sum_{l=1}^{L} F1_l \tag{7}$$

where $L$ is the total number of labels and $F1_l$ is the F1 score of the $l$-th label.

Micro-F1 reflects overall predictive performance, but is typically influenced more strongly by frequent labels. By contrast, macro-F1 gives equal weight to each label and therefore better captures performance on rare classes. A large gap between micro-F1 and macro-F1 often indicates that the model performs substantially better on head labels than on tail labels under imbalanced distributions.

***Example-based F1*** evaluates prediction quality at the sample level by comparing the predicted label set with the ground-truth label set for each image. For sample $i$, precision and recall are defined as

$$P_i = \frac{|Y_i \cap \hat{Y}_i|}{|\hat{Y}_i|}, \qquad R_i = \frac{|Y_i \cap \hat{Y}_i|}{|Y_i|} \tag{8}$$

where $Y_i$ is the ground-truth label set of sample $i$, $\hat{Y}_i$ is the predicted label set, $|Y_i \cap \hat{Y}_i|$ is the number of correctly predicted labels, $|\hat{Y}_i|$ is the number of predicted labels, and $|Y_i|$ is the number of true labels.

The F1 score for sample $i$ is then computed as

$$F1_i = \frac{2P_iR_i}{P_i + R_i} \tag{9}$$

The final example-based F1 is obtained by averaging $F1_i$ over all $N$ samples.

Larger example-based F1 values indicate stronger agreement between predicted and ground-truth label sets at the individual-sample level. Lower values suggest that the model either omits relevant labels or introduces excessive false positives.

***Jaccard Index*** measures the similarity between the predicted and ground-truth label sets for each sample:

$$J_i = \frac{|Y_i \cap \hat{Y}_i|}{|Y_i \cup \hat{Y}_i|} \tag{10}$$

where $Y_i$ is the ground-truth label set of sample $i$, $\hat{Y}_i$ is the predicted label set, $|Y_i \cap \hat{Y}_i|$ denotes the number of shared labels, and $|Y_i \cup \hat{Y}_i|$ denotes the total number of distinct labels appearing in either set.

The final Jaccard Index is typically obtained by averaging $J_i$ over all samples. Larger values indicate greater overlap between predicted and ground-truth label sets, whereas lower values reflect larger mismatches due to missed labels, redundant predictions, or both.

***Precision@K*** and ***Recall@K*** evaluate the quality of the top-$K$ predicted labels. For sample $i$,

$$P@K_i = \frac{|TopK_i \cap Y_i|}{K}, \qquad R@K_i = \frac{|TopK_i \cap Y_i|}{|Y_i|} \tag{11}$$

where $TopK_i$ denotes the set of top-$K$ labels with the highest predicted confidence scores for sample $i$, $Y_i$ is the ground-truth label set, and $|TopK_i \cap Y_i|$ is the number of correct labels appearing in the top-$K$ predictions.

Larger Precision@K values indicate greater accuracy among the model's highest-ranked predictions, whereas larger Recall@K values indicate that a greater proportion of relevant labels is retrieved within the top-$K$ outputs. Low values on these metrics suggest limited ranking effectiveness in the most operationally relevant portion of the prediction list.

Overall, these metrics provide complementary perspectives on multi-label image tagging. Ranking-based measures emphasize confidence ordering, whereas label-based and example-based measures assess classification quality at different levels of granularity. In addition, head-tail analysis can further reveal robustness to label imbalance, the quality of semantic dependency modeling, and the contribution of feature-fusion components.

## 2.2 Data Imbalance Approach

### *2.2.1 The Nature of Imbalance in Multi-Label Image Tagging*

Data imbalance has long been recognized as a central challenge in machine learning, where classifiers trained on unequal distributions tend to favor majority classes while underperforming on minority ones [3]. Prior reviews consistently show that such imbalance not only distorts the learning process but also makes conventional evaluation metrics, especially overall accuracy, potentially misleading in practical tasks. In multi-label image tagging, however, this problem becomes more complex than in conventional single-label classification. Since each image may be associated with multiple semantic tags, the model must solve many highly asymmetric binary decisions simultaneously, with only a few positive labels and a large number of negatives for each instance. In large-scale datasets such as NUS-WIDE and MS-COCO, label frequencies often follow a long-tail distribution, where dominant labels are repeatedly observed while rare labels remain severely underrepresented [4]. This imbalance makes optimization increasingly biased toward frequent labels and weakens the learning for rare concepts, thereby limiting the model's ability to recognize minority classes reliably. Existing studies generally address imbalance through either data-level or algorithm-level strategies [3].

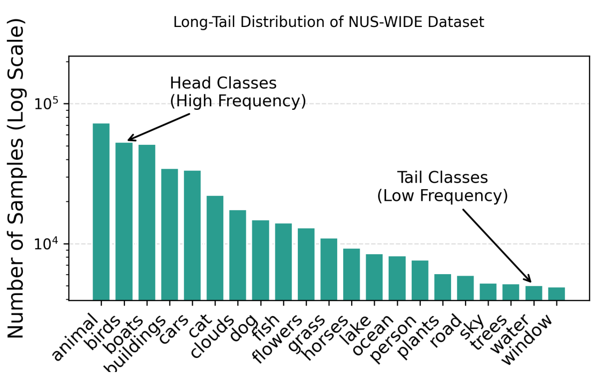


Fig. 2. Long-tail problem in NUS-WIDE

### *2.2.2 Data-level strategy*

Data-level strategies attempt to rebalance the training distribution before learning begins, mainly through oversampling and undersampling. Oversampling increases the representation of minority classes, either by duplicating existing samples or by generating synthetic examples through methods such as SMOTE and its variants, GAN-based methods and Active Balancing Mechanism (ABM) [3]. In contrast, undersampling reduces the size of the majority class by removing selected instances, with representative techniques including random undersampling, Tomek Links, and Cluster Centroids [3]. These methods are widely discussed in the broader imbalanced-learning literature because of their conceptual simplicity and practical effectiveness in many classification settings such as medical and engineering field [3]. However, their applicability becomes less straightforward in multi-label image tagging, where labels are not independent, features are often high-dimensional, and naive resampling may distort semantic structure or introduce additional noise.

### *2.2.3 Algorithm-level strategy*

Algorithm-level approaches instead modify the learning objective so that minority labels receive greater attention during training. A classical direction is cost-sensitive learning, where misclassification of minority classes is penalized more heavily [3]. In deep multi-label recognition, this idea has evolved into imbalance-aware loss design. Binary cross-entropy (BCE) serves as a standard baseline for deep multi-label classification, but under highly imbalanced label distributions it is often dominated by large numbers of easy negative labels. To address this limitation, Focal Loss extends BCE by shifting the learning focus toward harder samples [13]. Upon this idea, Asymmetric Loss further adapts the formulation specifically for multi-label settings to reduce the influence of abundant easy negatives and promoting more balanced optimization [14]. Wu et al. [15] explored distribution-balanced loss for long-tailed multi-label settings, suggesting that imbalance should be treated as a label-structure problem rather than a purely numerical one.

### *2.2.4 Limitations of Data Imbalance Approach*

Despite substantial progress, current imbalance-handling methods still face important limitations. First, most approaches operate at the loss level by reweighting positive and negative samples, but this does not necessarily improve the model's ability to learn the feature for rare labels. In multi-label image tagging, minority labels often correspond to less distinctive visual objects. If these patterns are not well captured by the feature extractor, simply increasing their loss weight may not help the model distinguish them from majority classes. The model may still confuse rare labels with more frequent ones, even when imbalance-aware objectives are applied [16]. Second, imbalance in multi-label tasks is closely intertwined with label dependency. Rare concepts often do not occur in isolation; instead, they co-occur with other labels in structured semantic patterns [15] .Methods that treat each label independently may therefore overlook valuable relational information that could support tail-label prediction. Recent work on co-occurrence-aware reranking and label-dependency modeling reflects this shift toward more structured treatments of long-tailed multi-label recognition [15].

## 2.3 Class Co-occurrence Approach

Approaches based on label co-occurrence information have emerged as a primary paradigm in addressing multi-label problems, replacing the traditional assumption of label independence [12], [17], [18]. Rather than decomposing a multi-label task into a collection of independent single-label classification problems, this approach focuses on modeling the probability of label occurrences based on the co-occurrence frequency between labels and concepts. Compared to traditional methods, leveraging co-occurrence information offers significant advantages. Specifically, it enables the model to capture the semantic dependencies existing between labels. For instance, labels such as 'mountain' and 'tree' exhibit a high frequency of simultaneous occurrence, whereas the label 'desert' has a substantially lower probability of appearing alongside 'tree'. Integrating these statistical insights enhances the model's internal logic and ensures more consistent predictions. Based on the complexity and the order of these relationships, existing methodologies are generally categorized into second-order and high-order strategies.

#### 2.3.1 Second-order Strategies

Second-order strategies focus on modeling pairwise correlations between labels, a direction widely adopted due to its favorable balance between predictive performance and computational efficiency. A seminal contribution in this domain is the Rank-SVM [19], which employs a ranking loss function to ensure that relevant labels are prioritized over irrelevant ones in the scoring space. Building upon this, Li et al. [17] proposes a smooth pairwise ranking loss function combined with an optimal confidence threshold estimation module to enhance multi-label image classification within deep learning frameworks. Another approach for second-order strategies is to leverage empirical statistics derived directly from training data. For instance, Pruned Set transforms the multi-label task into pruning strategy, where the model learn pruning away infrequently occurring label sets to preseve more frequently occurring set. Recently, there has been a shift toward integrating these statistical priors directly into deep learning frameworks. This integration typically follows two primary paths: prior initialization or post-refinement. While some studies perform informed weight initialization of the penultimate layer to establish semantic priors [18], a growing body of work focuses on refining initial estimates at the decision stage. This is exemplified by frameworks that employ Graph Convolutional Networks (GCNs) or statistical calibration modules to adjust raw logits based on historical co-occurrence frequencies [20], [21]. By enforcing these empirical conditional probabilities, such methods effectively align class representations with real-world patterns, facilitating faster convergence and enhanced predictive precision, particularly in sparse label spaces. However, pairwise concept co-occurrence can only depict the relationship between pairwise by cannot understand the global context due to pairwise constrain. For example, 'Rubber Gloves' or 'Glassware' may appear in various unrelated contexts like 'Cleaning' or 'Dining', yet their joint occurrence creates a unique high-order synergy that exclusively identifies a 'Scientific Experiment'."

#### 2.3.2 High-order Strategies

High-order strategies aim to capture intricate dependencies where a label's presence is conditioned on a subset or the entire set of remaining labels. A foundational framework in this category is Classifier Chains (CC) [22], which transforms the multi-label task into a sequential prediction problem, conditioning each classifier on all preceding label outputs. To address the inherent sensitivity to chain ordering, Probabilistic Classifier Chains (PCC) [23] were introduced to optimize the joint probability distribution across the label space.

Moving beyond fixed sequences, recent advancements in geometric deep learning have shifted the focus toward structural modeling. For instance, ML-GCN [24] leverages Graph Convolutional Networks to treat labels as nodes, propagating high-order dependency information across edges defined by correlation metrics. More recently, the field has embraced self-attention mechanisms to overcome the rigid topologies of graphs. Architectures such as C-Tran [12] utilize learnable label embeddings integrated with Transformer-based self-attention, enabling the model to dynamically capture global dependencies and high-order relationships across the entire label set without the constraints of predefined structures.

#### 2.3.3 Limitations of Co-occurrence Approach

While approaches leveraging co-occurrence information significantly enhance a model's ability to capture label dependencies, they are subject to several inherent limitations. A primary drawback arises from the fact that these statistics are typically extracted directly from the training data, which often exhibits a profound imbalance in category distribution [25].

Consequently, the derived co-occurrence information tends to be heavily biased toward frequent labels. This results in a "rich-get-richer" effect, where the representations of relationships between majority classes are strengthened, while the associations involving minority (tail) classes remain poorly represented or entirely neglected. Therefore, employing co-occurrence strategies without explicitly addressing the data-imbalance problem prevents the model from achieving a holistic understanding of the entire label space. Instead of generalizing across all categories, the model may become over-specialized in predicting common label clusters, further marginalizing the rare but critical categories within the dataset.

### 2.4 Data Fusion Approach

Data fusion aims to uniformly model visual features from various sources or levels to enhance the semantic representation capability of images. In complex multi-label image tagging tasks, a single visual feature often fails to capture the intricate topological structures of multiple co-occurring objects and their semantics [2], [7]. Multi-modal or multi-view data fusion mechanisms are therefore required to explicitly capture high-order semantic correlations among diverse labels [7]. While the broad effectiveness of such fusion strategies has been validated in specialized domains, including multisource data integration in remote sensing, their application in multi-label scenarios is specifically crucial for resolving visual ambiguities and aligning heterogeneous features with complex label spaces. Based on the stage of feature interaction, data fusion methods can be categorized into early fusion and deep fusion [2].

#### 2.4.1 Early Fusion

Early fusion strategies aim to integrate multiple modalities or heterogeneous features at the initial or intermediate stages of a model, prior to task-specific high-level semantic abstraction. In the context of multi-label tagging, traditionally, this approach focused on original raw or low-level features directly at the input stage to construct a joint representation [26], enabling the model to exploit cross-source complementarities—such as linking object textures with global scene contexts—from the onset of the learning process.

Recently, there has been a significant shift toward expanding the scope and methodology of early fusion to better support highly correlated label spaces. Instead of merely combining low-level feature maps, modern architectures increasingly project diverse modalities—such as visual, textual, and auditory data—into a shared representational space. A prominent advancement in this domain involves token-based early-fusion foundation models, which quantize different modalities into discrete tokens. This allows a unified transformer architecture to seamlessly process interleaved sequences without relying on separate, modality-specific encoders [27]. By integrating these representations early, such models facilitate powerful joint reasoning, allowing the model to simultaneously weigh visual evidence and semantic label dependencies right from the initial layers.

Furthermore, contemporary early fusion is no longer strictly confined to the absolute input layer; it has evolved to encompass deep interactive fusion at intermediate encoder stages. By employing cross-aware early fusion mechanisms, frameworks can enable vision and language features to bidirectionally reference each other's information across various intermediate layers [28]. This mutual interaction significantly enhances the robustness of both encoders and improves cross-modal context modeling, proving particularly effective for disambiguating overlapping objects in multi-label scenes compared to relying solely on high-level feature alignment. To address the immense computational overhead associated with processing mixed modalities simultaneously from early stages, recent studies have integrated modality-aware mixture-of-experts (MoE) architectures into early-fusion models. These structures allocate specific expert parameters to different modalities, maintaining the semantic adaptivity necessary for diverse label sets and training efficiency while preserving the fundamental benefits of early integration [29].

Despite these profound advantages in seamless cross-modal integration, early fusion inherently faces distinct technical challenges in multi-label systems. As demonstrated by Baltrušaitis et al. [2], simple early concatenation exhibits a high sensitivity to feature dimensional inconsistency and noise redundancy, which can easily mislead multi-label classifiers. Moreover, optimizing advanced shared-space early fusion architecture necessitates specialized training techniques and architectural modifications to ensure stability, relieve modality imbalance, and effectively bridge the heterogeneity gap between fundamentally disparate data sources [27].

#### 2.4.2 Deep Fusion

To overcome the limitations of early fusion in handling highly abstracted concepts, deep fusion operates in learned feature spaces and leverages deep neural networks for advanced representation learning. It utilizes attention mechanisms, Graph Neural Networks (GNNs), or cross-modal Transformers to achieve semantic alignment across heterogeneous features, which is particularly vital for mapping complex visual patterns to specific label embeddings.

The specific advantages of deep fusion lie in its capacity for dynamic alignment and global semantic modeling. For instance, Guo et al. [26] demonstrate that attention mechanisms can adaptively assign semantic weights across heterogeneous features, significantly improving feature alignment and mitigating noise interference. Furthermore, recent frameworks map diverse features into graph structures or sequences to explicitly capture global semantic interactions within deep representations [30]. These approaches enable the adaptive weighting of feature channels to highlight regions relevant to specific tags, the propagation of relational information across the label co-occurrence graph, and the modeling of high-order semantic interactions. As a result, deep fusion provides a flexible and robust integration of heterogeneous information, making it a highly effective strategy in complex multi-label systems.

#### 2.4.3 Limitations of Data Fusion Approach

Despite these advancements, several challenges remain in large-scale multi-label settings. The heterogeneity gap between features can lead to semantic misalignment [26]. Deep fusion modules with large parameter spaces increase the risk of overfitting when high-quality labels are limited [30]. Crucially, in long-tailed datasets, joint fusion strategies are often dominated by frequent labels, leading to insufficient representation for rare labels. In addition, existing interaction modules rely on relatively shallow mappings, limiting their ability to capture complex spatial relationships and high-order semantic dependencies [7], [26].

## 3 Research Directions

### 3.1 Research Gaps

While the aforementioned strategies have made significant contributions to the field of multi-label image tagging, they are currently characterized by a major limitation: they are predominantly applied in isolation. As previously discussed, real-world multi-label problems rarely stem from a single issue like data imbalance or semantic entanglement; instead, they arise from a complex combination of these factors.

Specifically, existing approaches exhibit the following gaps:

***Data-imbalance approaches***: These methods typically address the disproportionate distribution of labels by treating them as independent entities. In doing so, they tend to overlook the intrinsic semantic relationships and dependencies between labels.

***Co-occurrence and semantic approaches***: Conversely, these strategies focus on representing the semantic correlations between labels. However, such relationships often fail to be effective when the underlying data suffers from extreme bias or imbalance. If the statistical representation is heavily skewed, the model's ability to learn accurate label dependencies is significantly compromised.

***Data-fusion approaches***: These modules often face semantic misalignment between heterogeneous features and high overfitting risks when labels are scarce. In long-tailed scenarios, fusion strategies are typically dominated by frequent labels, resulting in poor representations for rare classes. Furthermore, current methods rely on shallow mappings, failing to capture complex spatial and high-order semantic dependencies.

***Evaluation metrics***: Standard measures such as mAP, F1-score, and the Jaccard Index fail to properly assess a model's semantic understanding due to their binary nature. These metrics penalize all mismatches equally, regardless of their semantic proximity. For instance, if a model predicts "water" for an image labeled as "ocean", these metrics treat it as a complete failure, ignoring the fact that the prediction is semantically consistent with the context.

The primary gap in the current literature is the absence of an integrated framework that addresses these challenges simultaneously. There is a clear need for a holistic approach that can manage data imbalance while maintaining the integrity of label semantic correlations and providing a more nuanced evaluation of semantic reasoning, as these problems are fundamentally interconnected in large-scale image tagging tasks.

### 3.2 Proposed Integration

To bridge the identified gaps, we propose a holistic framework that synergizes multimodal feature fusion, semantic correlation modeling, and robust imbalance handling into a unified architecture. Specifically, our approach first integrates heterogeneous features to create a rich representation base, which is then refined by incorporating co-occurrence and semantic information to capture high-order label dependencies. To mitigate the effects of long-tailed distributions, we integrate data-imbalance handling techniques—such as advanced data augmentation or cost-sensitive loss functions—directly into the learning process.

Crucially, this integration extends beyond the training phase to address the limitations of traditional evaluation. We propose a novel evaluation methodology that shifts from binary label matching to semantic-based performance assessment. By incorporating semantic proximity measures, this method will validate the model's performance based on its ability to maintain contextual consistency—recognizing, for example, that predicting "water" for an "ocean" label reflects a higher level of semantic reasoning than an unrelated error. This integrated framework aims to provide both a more robust tagging system and a more accurate measure of true model intelligence in large-scale scenarios.

## 4 Conclusion

This literature review has examined multi-label image tagging through three tightly coupled perspectives: data imbalance, semantic dependency modeling, and data fusion. A central finding is that current research largely treats these components as isolated problems, limiting the effectiveness of existing solutions — rare labels in particular remain difficult to predict, as they are simultaneously affected by weak feature representations, insufficient statistical support, and poorly modeled semantic relationships. More fundamentally, data imbalance, label dependency, and feature representation are not independent challenges but inherently interconnected aspects of a unified learning problem. Current evaluation protocols further compound this issue, as commonly used metrics (e.g., mAP, F1-score, Jaccard Index) fail to capture semantic proximity or contextual consistency between labels. Future research should therefore move toward unified, structurally aware frameworks that jointly model imbalance and dependency, integrate feature fusion with semantic reasoning, and adopt evaluation metrics that better reflect semantic alignment. Achieving robust performance in real-world scenarios requires a shift from fragmented solutions toward integrated approaches — and this review provides a structured perspective on these interconnected challenges to guide that direction.

## Appendix A
## AI Use Statement

Generative AI tools (e.g., ChatGPT) were used in a limited and supportive capacity during the preparation of this literature survey. Specifically, these tools were employed for language polishing, improving clarity and grammar, and assisting with the organisation of ideas.

All literature search, paper selection, reading, interpretation, and synthesis were conducted independently by the authors. No AI tools were used to generate literature summaries or to replace the authors' own critical analysis. All claims, comparisons, and citations included in this work have been verified directly from the original sources.

The final manuscript reflects the authors' own understanding and academic judgement.

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